



Course: Data Visualization & Analytics

Assignment 2: Advanced Interactive Dashboards via D3

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Table of Contents

[Advanced D3.js Interactive Dashboards: Technical Report](#)

Summary

Task 1: Spatial & Network Analysis

Graph 1: Zoomable Aggregate Map

Graph 2: Fuel Force-Directed Bubble Chart

Graph 3: Brushable Capacity Timeline

Task 2: Temporal & Hierarchical Simulation

Overview

Graph 4: Motion Chart (GDP vs. Life Expectancy)

Graph 5: Life Expectancy Choropleth Map

Graph 6: Hierarchical Donut Chart

Cross-Task Integration

I used a controller pattern to coordinate filters and interactions across all components: The Task 1 controller handles fuel selection and year range updates to both map and bubble chart. Task 2 shares state across the motion chart, choropleth, and hierarchy.

SVG elements are made responsive via the responsivefy() helper by scaling responsively for different screen-sized platforms. Also, Promise.all() makes data loading asynchronous, making sure that the loading of the dashboards happens fast without freezing up the browser.

Conclusion

Assignment 2: Advanced Interactive Dashboards via D3

Summary

Within this project, I have developed two interactive dashboards using D3.js v7. My aim was to create intuitive and visually attractive means of exploring complex datasets interactively. Both of the dashboards incorporate various advanced techniques in semantic zoom, cross-filtering, temporal animation, and hierarchical data visualization to enable the user to dig into the data from multiple angles.

The first dashboard shows data about power production around the world, both at a country level and at a plant level. The second one is inspired by Gapminder; it visualizes socioeconomic indicators across time, including motion charts, choropleth maps, and hierarchical donut charts. In this report, I walk through each step and explain my decisions, showing how the workflow would look with images at each stage.

Task 1: Spatial & Network Analysis

Task 1 is all about understanding the global power plant data from the World Resources Institute. I have combined three components in a single dashboard.

- A zoomable map that toggles between country-level aggregates and individual plants.
- A force-directed bubble chart to explore fuel types.
- A brushable timeline for filtering the data by year.

All three components interact to provide a means for users to explore the dataset from different angles simultaneously.

Graph 1: Zoomable Aggregate Map

I loaded the power plant CSV and the world GeoJSON simultaneously by using `Promise.all()`. This prepared the data: normalizing field names, converting the capacity to numbers, and calculating the centroids for each country. This preparation made sure that the data was clean and ready for visualization.

Projection for the basemap: `geoNaturalEarth1`. Fit projection to SVG size. Draw countries with a neutral grey fill and white borders so that it gives geographic context but doesn't compete with the data.

It uses semantic zoom so that above a certain zoom level, the map switches from displaying country-level aggregates to the individual plants. There are two separate layers that control which circles are visible: `aggregateLayer` and `detailLayer`. The radius of the circles will be dynamically adjusted to ensure clarity.

The aggregate layer groups plants by country and sums capacities using `d3.rollups()`. The Circle radii are scaled using a square root scale to balance representation, and semi-transparent blue fills let the underlying map remain visible.

Zoomable Aggregate Map

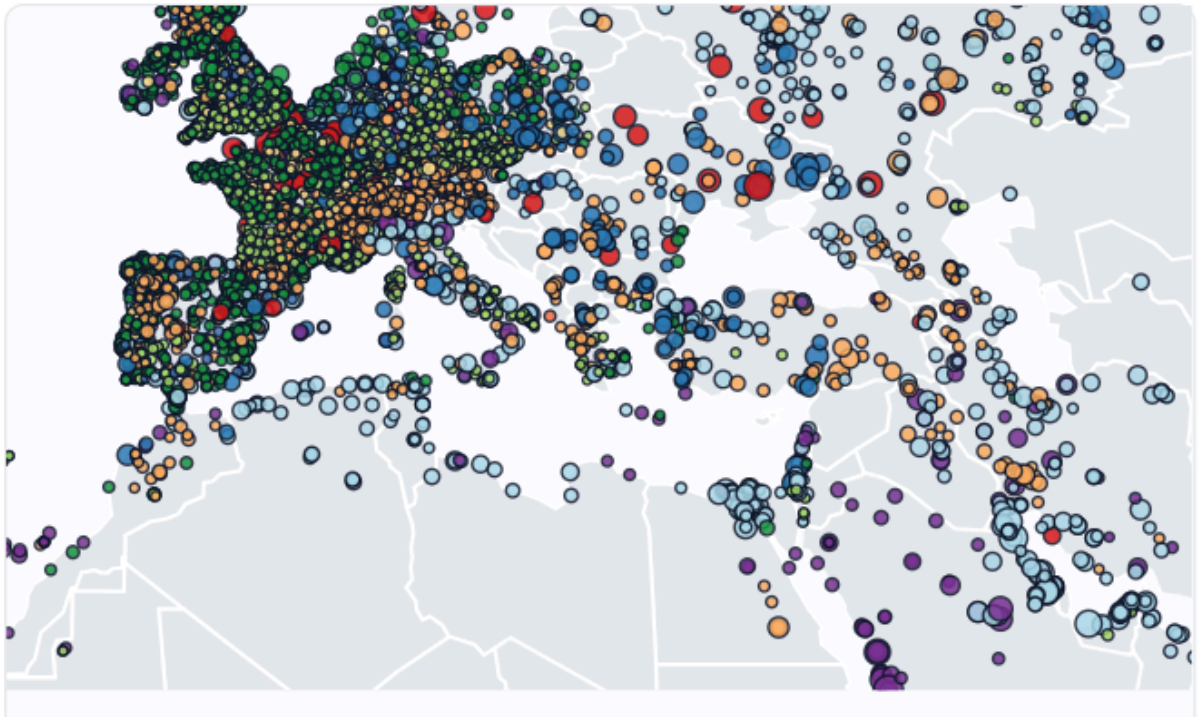


Plants: 17,322 **Capacity:** 3.7M MW

Individual plants are filtered for valid coordinates in the detail layer and circles proportional to their capacity are drawn on it. Color encodes fuel type, guiding users to fast identification of

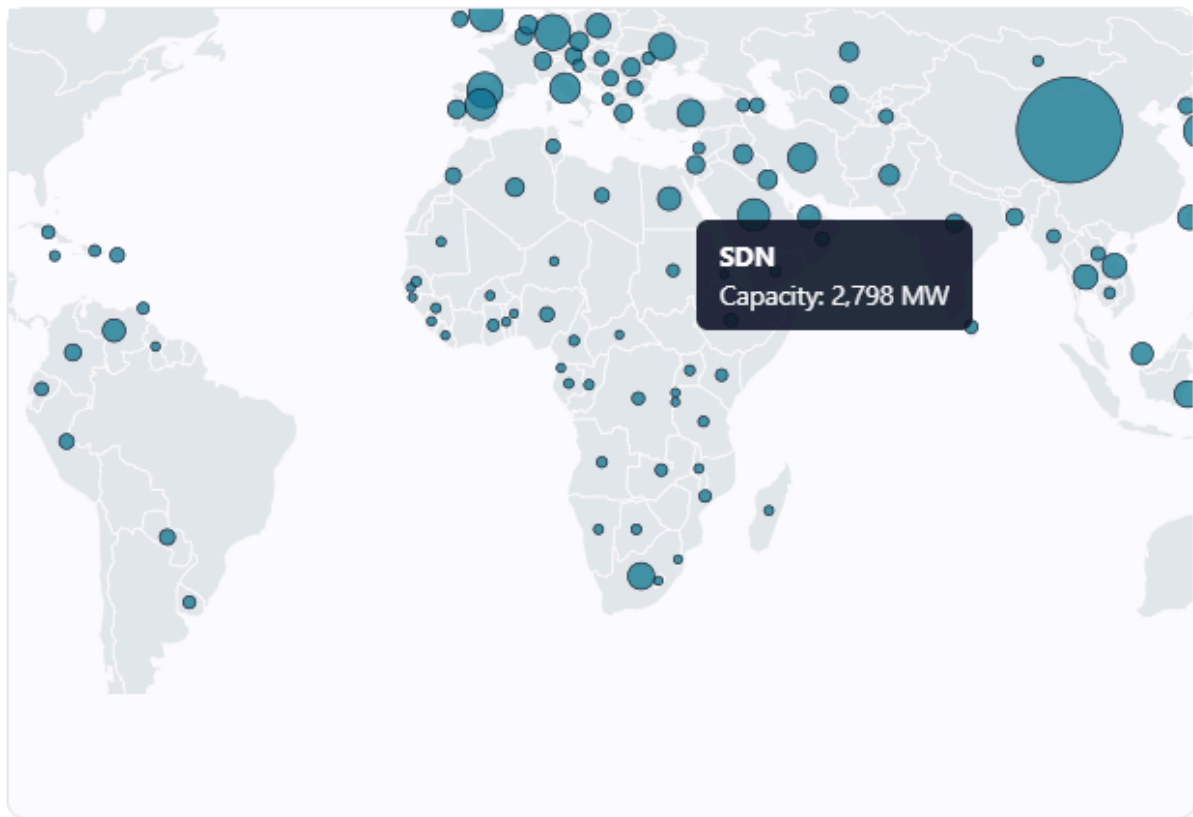
patterns across regions.

Zoomable Aggregate Map



Tooltips for both layers have been added, with total capacity for countries or detailed information for plants, following the mouse cursor to show context without cluttering the visualization.

Zoomable Aggregate Map



Plants: 17,322 **Capacity:** 3.7M MW

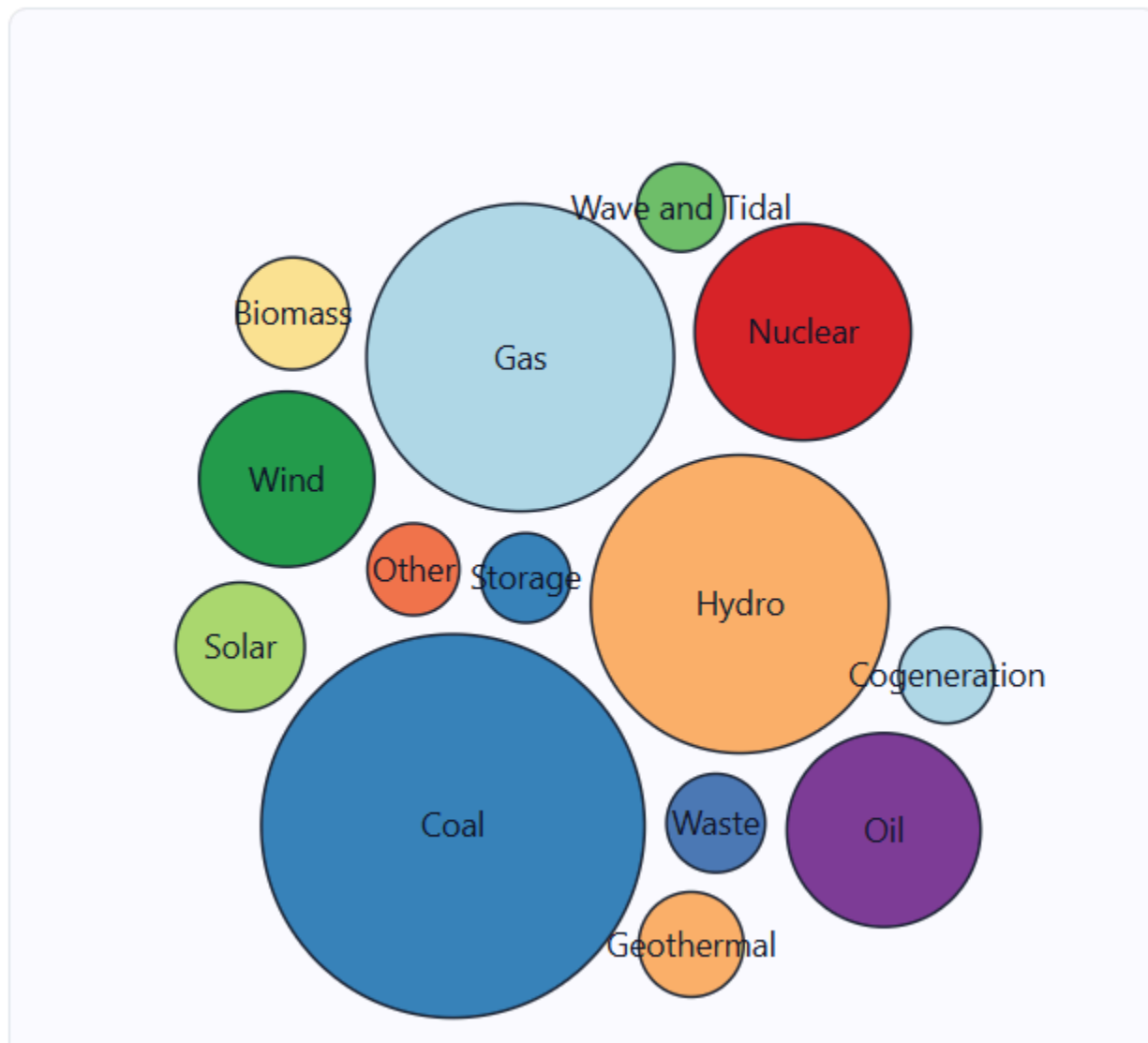
Graph 2: Fuel Force-Directed Bubble Chart

I set up a force simulation on the fuel-type bubbles, applying repulsion to avoid overlap and weak centering forces to maintain node clumps, allowing the bubbles to lay out organically and dynamically adjust as data changed.

Data aggregation was performed by fuel type, determining total capacity and count of plants. Scaling of bubble radius used square root scale, while colors were assigned to provide contrast between fuels; labeling of text was placed centered at each bubble. Clicking on a bubble

highlighted that fuel type, filtering the map accordingly.

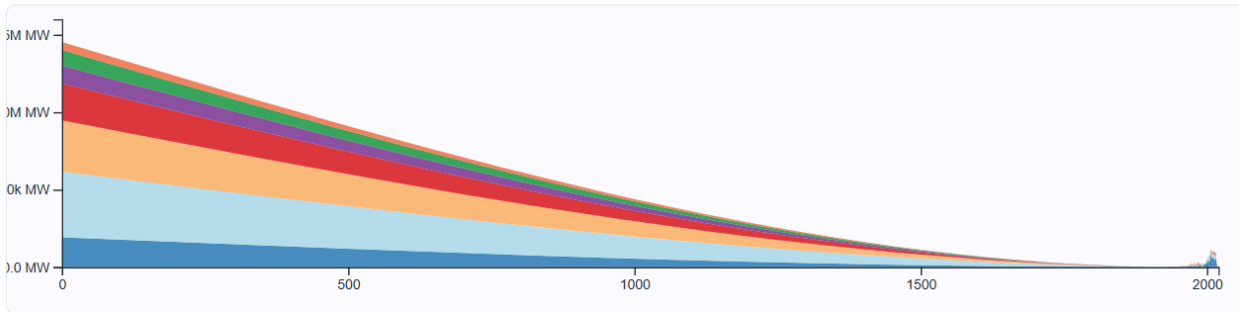
Fuel Force Bubbles



Graph 3: Brushable Capacity Timeline

I designed a stacked area chart of capacity over time by fuel type. I separated out the top six fuels and combined all other fuels into an "Other" category. Via a brush, users can select a year range to update the map and bubble chart with only relevant data. Years are shown along the x-axis with readable tick intervals. Capacity is displayed on the y-axis with compact formatting, i.e., "1.5M MW". Transitions are smooth to keep continuity across filtering, making changes easy to track visually.

Brushable Capacity Timeline



Task 2: Temporal & Hierarchical Simulation

Overview

Task 2 focuses on indicators of socioeconomic development, namely GDP per capita, life expectancy, and population over time. Guided by Gapminder, this should be done in three components simultaneously:

- A motion chart of GDP vs. life expectancy with circle size representing population
- A choropleth map showing life expectancy by country.
- A donut chart showing a hierarchical population distribution

These are synchronized by a time slider and controls for play/pause.

Graph 4: Motion Chart (GDP vs. Life Expectancy)

Then I joined several CSV files of GDP, life expectancy and population, combining these sources by country and year. I kept only countries that had all three values and sorted the dataset into a Map keyed by year so that animation can be smooth.

GDP per capita was mapped to the x-axis using a logarithmic scale to handle the wide range of values. Population values were mapped to circle radii using a square root scale, and countries were colored by continent with a distinct, colorblind-friendly palette.

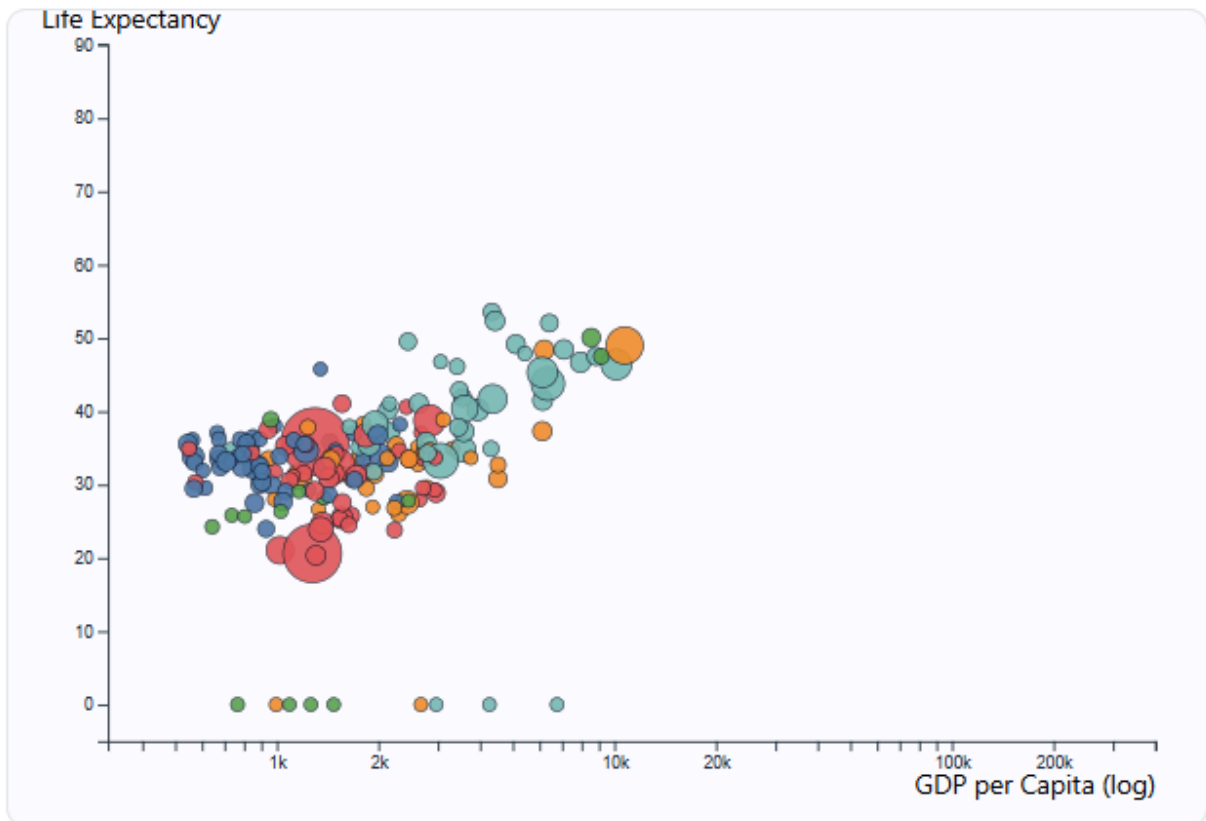
A log scale was used to map the great range of GDP per capita values to the x axis. Values of population were mapped to circle radii via a square root scale. Countries were colored by continent using a colorblind-friendly unique palette.

Animations were achieved with `d3.interval()`; in it, the year was updated every 1.5 seconds and transitions also took 600ms. The enter/update/exit pattern makes for smooth transitions when countries appear, move, or disappear over time.

Play

Year: 1900

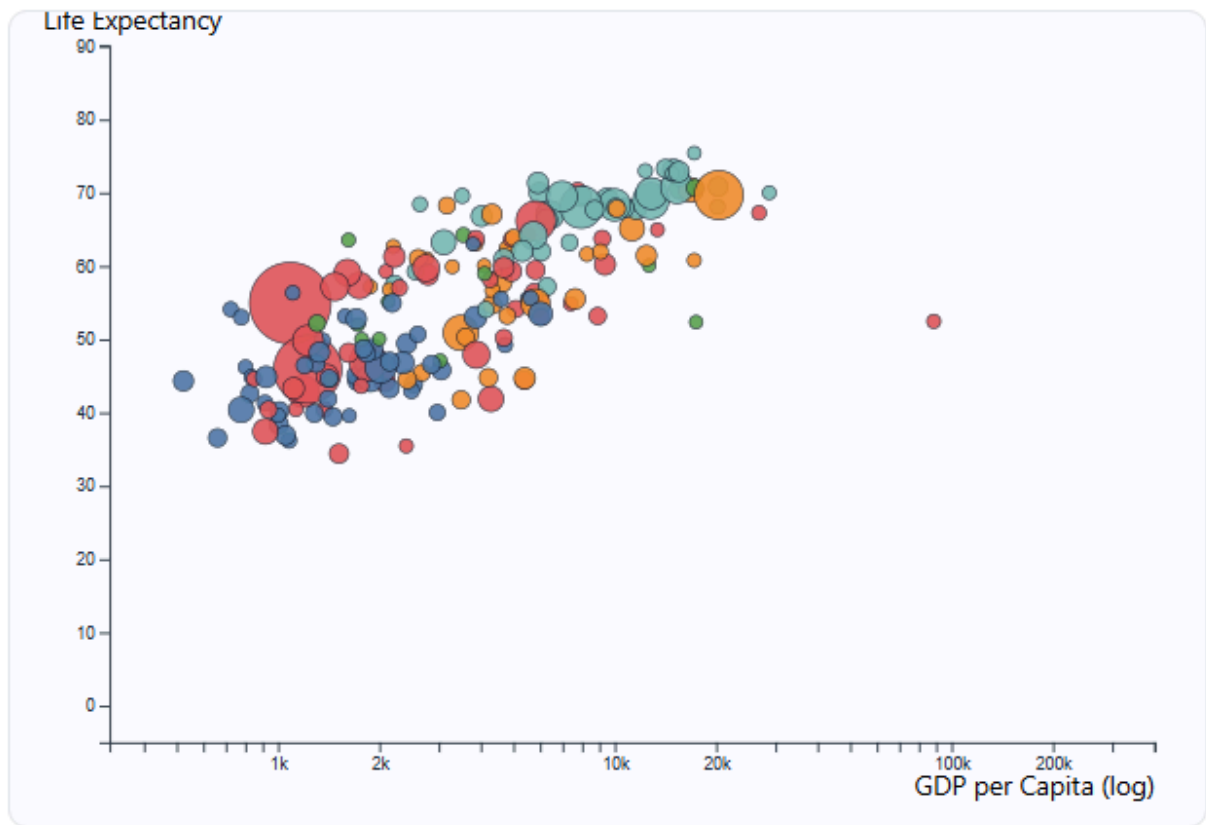
Motion Chart



Play

Year: 1957

Motion Chart



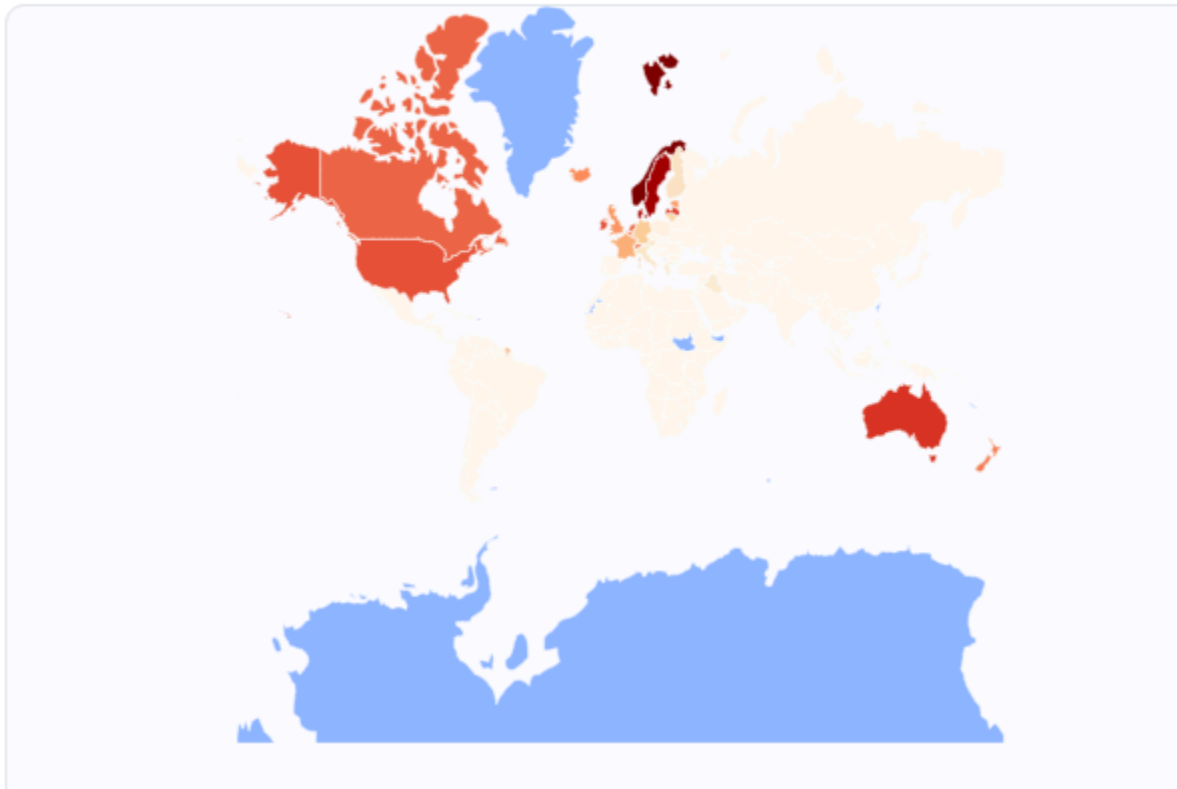
Graph 5: Life Expectancy Choropleth Map

The choropleth uses a Mercator projection for sizing to fit the map to the size of the SVG. By default, the countries are gray, and there is a life expectancy encoding using a sequential orange to red color scale. The colors transition in unison with the movement of the year on the chart; there are tooltips too that show precise values on hover.

Play

Year: 1900

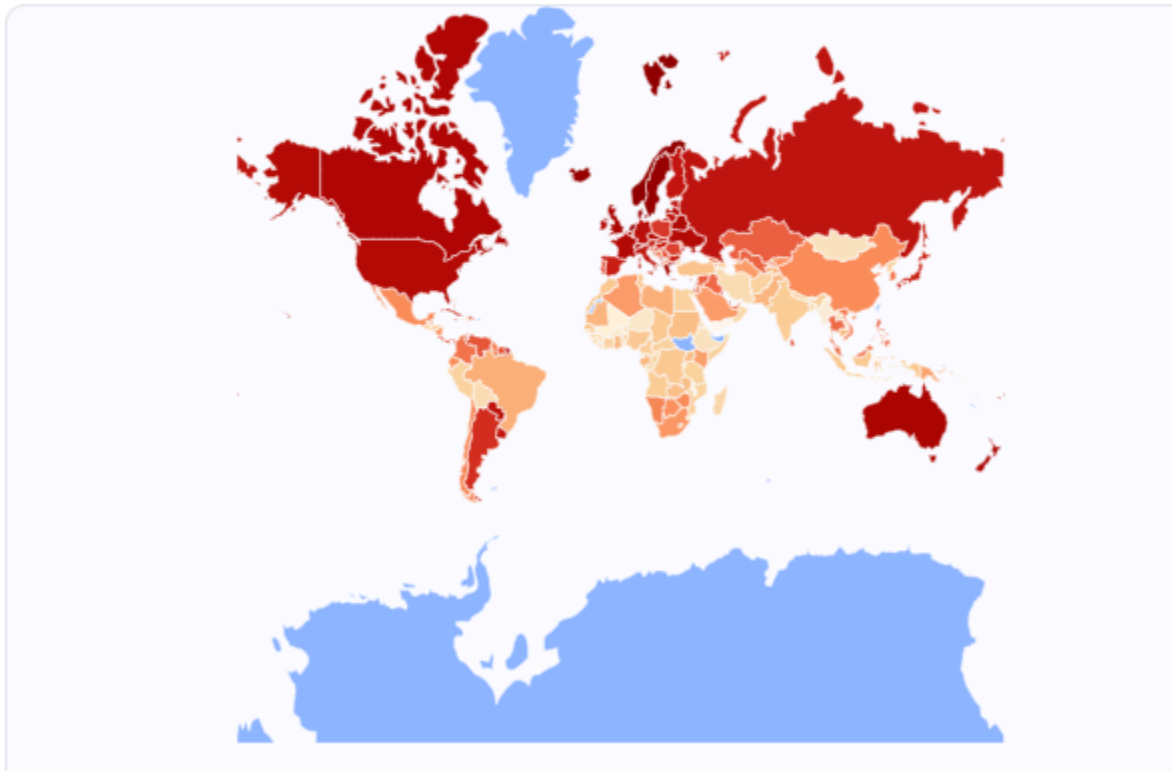
Life Expectancy Choropleth



Play

Year: 1957

Life Expectancy Choropleth



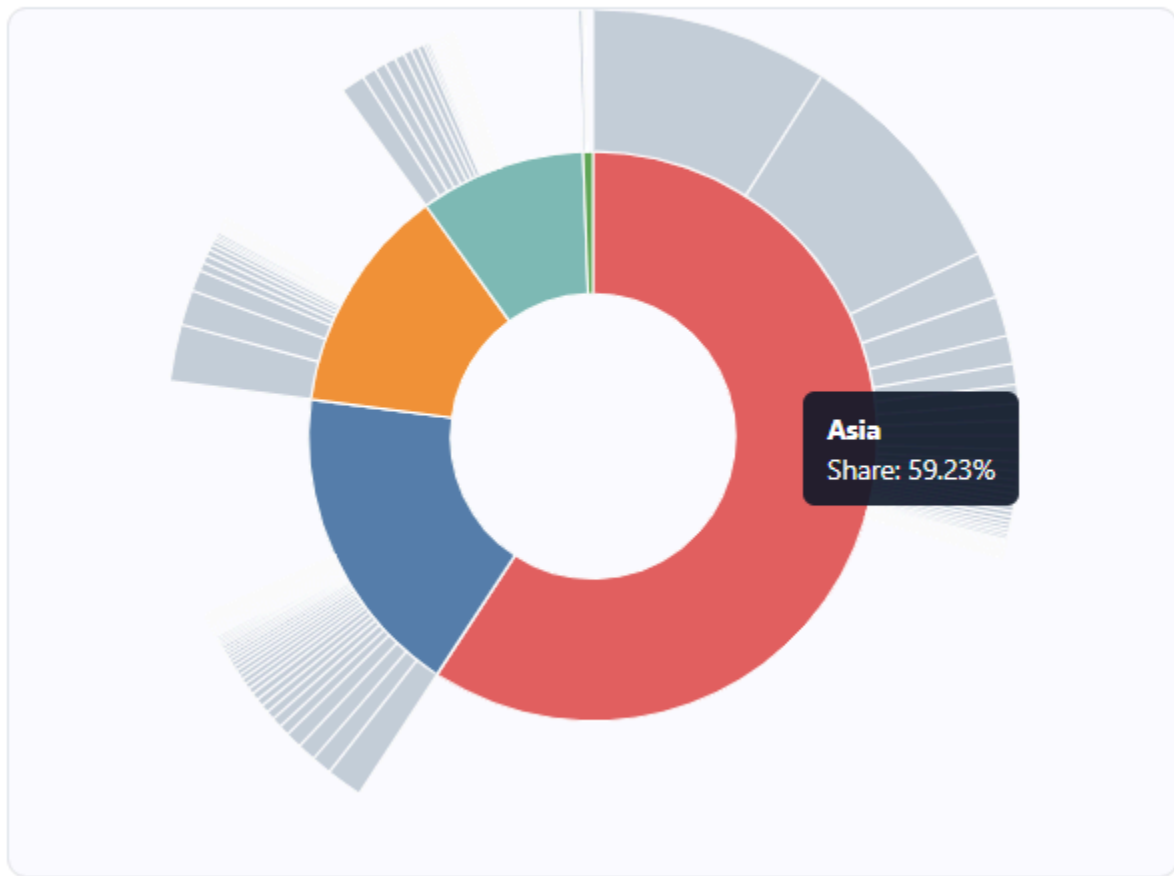
Graph 6: Hierarchical Donut Chart

The population data was transformed into a three-level hierarchy: World → Continent → Country. The computed partition layout calculated angular and radial positions for every segment. Different coloring separated arcs at the continent level, while on the country level, gray is used to represent neutrality, thus providing a visual hierarchy.

Hovering over the continents highlights the respective countries in the motion chart. Tooltips include population share percentages for continents and countries, adding quantitative context

with the visual hierarchy.

World → Continent → Country Hierarchy



Cross-Task Integration

I used a controller pattern to coordinate filters and interactions across all components: The Task 1 controller handles fuel selection and year range updates to both map and bubble chart. Task 2 shares state across the motion chart, choropleth, and hierarchy.

SVG elements are made responsive via the `responsivefy()` helper by scaling responsively for different screen-sized platforms. Also, `Promise.all()` makes data loading asynchronous, making sure that the loading of the dashboards happens fast without freezing up the browser.

Conclusion

This project shows how D3.js can create rich, interactive visualizations that let users explore complex datasets from multiple angles: semantic zoom, force-directed layouts, brushing, temporal animation, and hierarchical visualizations all combine to provide insight into global patterns of energy use and socioeconomic development. Coordinated interactions and cross-filtering transform static data into a dynamic exploration tool, letting users see patterns, trends, and relationships that would otherwise remain hidden.